

Rashmi Priya

Physicist — soft & active matter, glasses and amorphous solids, computational physics

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RESEARCH INTERESTS

Soft and active matter; physics of glasses and amorphous solids; yielding, plasticity and memory in disordered materials; liquid crystals; computational physics and large-scale molecular dynamics simulation.

EDUCATION

Ph.D. in Physics

Tata Institute of Fundamental Research, Hyderabad · Oct 2020 – Aug 2026
Advisor: Prof. Smarajit Karmakar

M.Sc. in Physics

Ramjas College, University of Delhi, New Delhi · 2017 – 2019

B.Sc. (Hons.) in Physics

Acharya Narendra Dev College, University of Delhi, New Delhi · 2014 – 2017

PUBLICATIONS

Tunable yielding and emergent rheology in amorphous solids with active particle doping

Rashmi Priya, Jürgen Horbach, Smarajit Karmakar
Nature Communications (2026) <https://doi.org/10.1038/s41467-026-75709-y>

Inverse Bauschinger to Bauschinger crossover under steady shear in amorphous solids — *Editors' Suggestion*

Rashmi Priya, Smarajit Karmakar
Physical Review Materials **10**, 035604 (2026) <https://doi.org/10.1103/stkn-vhrr>

Inverse Bauschinger effect in active ultrastable glasses — *Editors' Suggestion*

Rashmi Priya, Smarajit Karmakar
Physical Review E **112**, 035414 (2025) <https://doi.org/10.1103/4ws8-1s7j>

RESEARCH VISITS

Visiting Researcher, Heinrich Heine University, Düsseldorf, Germany

Hosted by Prof. Jürgen Horbach · Apr 2023 – Jun 2023

TALKS AND SEMINARS

- Soft Matter Group seminar, Heinrich Heine University (HHU), Düsseldorf, Germany, June 2023. Hosted by Prof. Jürgen Horbach.

SELECTED MEDIA COVERAGE

- EurekAlert!* (AAAS), August 2026.

FELLOWSHIPS AND EXAMINATIONS

- CSIR–UGC NET with Junior Research Fellowship (2019)
- Joint Entrance Screening Test (JEST) and GATE (2020); Joint Admission Test for M.Sc. (JAM) (2017)

CONFERENCES AND SCHOOLS

- HydSoft, University of Hyderabad (2025)
- Physics of Complex Systems, TIFR Hyderabad (2024)
- Soft and Living Matter: From Fundamental Concepts to New Material Design, ICTS (2023)
- Frontiers in Active and Soft Matter, TIFR Hyderabad and University of Hyderabad (2023)
- Celebrating the Science of Giorgio Parisi, ICTS (2021)

COMPUTATIONAL SKILLS

- Programming:** C (proficient)
- Parallel computing:** CPU-based parallel codes in C using MPI; large-scale simulation on high-performance computing clusters
- Simulation:** LAMMPS; custom molecular dynamics codes